eventSubscription

Service Description

Service ID: *"eventSubscription"*

Abstract

This document describes the `eventSubscription` service, version 1.0.0, produced by the EventHandler support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `eventSubscription` service, which provides subscription of the requester to an event type, with optional metadata filter and validity window. It is produced by the EventHandler system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Successor of the Arrowhead 4.6 EventHandler subscription services.

1.2 How This Service Is Meant to Be Used

A subscriber calls `subscribe` with the event type and an HTTPS `notifyUri`, and `unsubscribe` with the subscription identifier.

1.3 Important Delimitations and Dependencies

With TLS the `notifyUri` must use HTTPS. A system can only manage its own subscriptions here.

2 Service Interface

This section describes the interfaces to the `eventSubscription` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.



Figure 1: SysML-style block of the `eventSubscription` service and its operations.

The following interface operations are available.

2.1 operation subscribe (SubscriptionRequest) : SubscriptionResponse

Creates a subscription of the requester.

2.2 operation unsubscribe (Identifier)

Removes a subscription of the requester. Returns 200 if removed, 204 if it did not exist.

3 Information Model

Here, all data objects that can be part of the `eventSubscription` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.3, which are used to represent things like hashes and identifiers.

3.1 struct SubscriptionRequest

A subscription to an event type.

Field	Type	Description
eventType	String	Event type, camelCase.
notifyUri	String	HTTPS endpoint to which matching events are posted.
matchMetadata	Boolean	Whether only events whose metadata contain `metadataFilter` are delivered.
metadataFilter	Map<String, String>	Metadata an event must contain to be delivered.
startsAt	String	Start of the subscription's validity window.
expiresAt	String	End of the validity window.

3.2 struct SubscriptionResponse

A stored subscription.

Field	Type	Description
id	Number	Identifier of the subscription.
subscriberSystemName	String	System that owns the subscription.
eventType	String	Event type, camelCase.
notifyUri	String	HTTPS endpoint to which matching events are posted.
matchMetadata	Boolean	Whether only events whose metadata contain `metadataFilter` are delivered.
metadataFilter	Map<String, String>	Metadata an event must contain to be delivered.
startsAt	String	Start of the subscription's validity window.
expiresAt	String	End of the validity window.
createdAt	String	UTC time of creation.
updatedAt	String	UTC time of the last change.

3.3 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
Map	A set of key-value pairs.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “EventHandler – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/EventHandler/documentation/v5.2.2/EventHandler-v5.2.2-SysD.pdf.

[2] ——, “eventSubscription HTTP/TLS/JSON – Interface Design Description (EventHandler),” Eclipse Arrowhead, IDD 5.2.2, 2026, eventHandler-v5.2.2-eventSubscription-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	